

OMNIPRESENT

Autonomous Cross-Functional Employee Onboarding & Orchestration Platform

Executive Overview, Technical Architecture & System Reference Manual

Target Audience: Leadership, Engineering, HR & SecOps	System Version: v1.0.0 (Production)	Date: September 2026
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1. Executive Overview & The High-Level Metaphor

Modern enterprises waste hundreds of hours per hire manually coordinating IT hardware, SaaS permissions, security compliance, background checks, and manager introductions across fragmented communication channels. The fundamental root cause is that existing **HRIS platforms are Systems of Record, not Systems of Execution**.

The Airport Control Tower & The Filing Cabinet Analogy:

- **Traditional HRIS (Workday, BambooHR, Rippling):** The filing cabinet in the hangar. It accurately stores flight manifests, passenger names, and passport numbers, but it cannot guide airplanes onto the runway.
- **Omnipresent:** The Air Traffic Control Tower. It continuously monitors the airspace, anticipates conflicts, orchestrates the ground crew, alerts the pilots, and ensures a seamless landing for every new hire on Day One.

2. How Omnipresent Acts on Top of Existing HRIS

Omnipresent does **not** replace your HRIS. Instead, it operates non-invasively above it as an autonomous orchestration fabric:

- **Inbound Webhook Listeners:** Exposes secure HTTP endpoints that receive candidate 'Hired' notifications in real time from Workday, BambooHR, Rippling, or custom webhooks.
- **Schema Normalization:** Translates disparate vendor-specific payloads into a unified canonical employee data model (name, role, level, department, location, clearance, tech stack).
- **SHA-256 Idempotency Engine:** Hashes incoming payloads to guarantee that network retries or duplicate webhook firings are acknowledged safely without duplicating work queues.
- **Cross-Departmental Dispatch:** Distributes work out of HR's hands into dedicated operational inboxes for IT, SecOps, Workplace Facilities, and Hiring Managers.
- **Bi-Directional Status Write-back:** Streams real-time progress, document verification links, and SOC2 compliance audit logs back into the central HRIS employee profile.

Feature / Capability	Existing HRIS (System of Record)	Omnipresent (System of Execution)
Core Mission	Store payroll, benefits, contract metadata	Coordinate cross-functional actions & eliminate Day-1 blockers
Onboarding Plans	Static, generic PDF checklist for everyone	Role-tailored dynamic DAG with parallel dependency queues
IT & Security	Blind to laptop shipping, Okta SSO, GitHub access	Directly tracks provisioning status & blocker escalations
Employee Support	Static FAQs or human HR ticketing backlog	State-Aware Copilot + Hybrid RAG policy answers (0 tokens)

Compliance Auditing	Disconnected spreadsheets and Slack threads	Cryptographically sealed append-only audit trail (SOC2 ready)
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3. What AI is Actually Doing in This System

Rather than treating AI as a gimmicky chatbot wrapper, Omnipresent integrates AI selectively where probabilistic reasoning and contextual synthesis deliver exponential leverage over static rules:

AI Subsystem	Underlying Architecture	Operational Function in Omnipresent
1. Dynamic Plan Compiler	Azure OpenAI (gpt-5-mini) with Structured Output Schema	Analyzes candidate role, clearance, seniority, and tech stack to generate a custom 15-20 task Directed Acyclic Graph (DAG) with SLA estimates and parallel queues.
2. Hybrid RAG Policy Engine	MiniSearch BM25 Lexical + Subword Cosine k-NN Dense Vectors	Searches enterprise handbook policies (stipends, equipment, PTO, security). Returns authoritative answers with exact citations and similarity confidence scores.
3. State-Aware Live Copilot	Deterministic Live Graph Resolver + In-Memory Store	Resolves real-time questions ('What tasks are assigned to me?', 'Who is blocking my laptop?') directly from the live database consuming ZERO LLM tokens .
4. Blocker Triage & Escalation	Contextual Root Cause Reasoner & SLA Monitor	When a task is blocked (e.g. 'Okta license pool empty'), categorizes the blocker, flags the responsible department, and suggests proactive remediation.

4. The End-to-End Operational Workflow

Step 1: Offer Signed & Inbound Ingestion

The candidate signs their offer in Workday/BambooHR. An inbound webhook triggers Omnipresent, which normalizes the payload, calculates an idempotency hash, and registers the onboarding in DRAFT state.

Step 2: Autonomous AI Plan Compilation

HR clicks 'Generate AI Plan'. Azure OpenAI compiles a role-tailored task DAG (e.g. 19 parallel tasks spanning IT, HR, Manager, and New Hire). Tasks include SLA hours and prerequisite dependencies.

Step 3: Human-in-the-Loop Approval & Queue Dispatch

HR reviews the synthesized plan and clicks 'Approve & Dispatch'. The onboarding status transitions to ACTIVE, and tasks immediately route into personalized inboxes.

Step 4: Parallel Departmental Execution & Blocker Remediation

IT provisions laptops and SSO; Managers schedule 1-on-1s; New Hires complete I-9 forms. If an item is blocked, it generates an immutable audit entry and alerts the relevant inbox.

Step 5: Automated Verification & Day-One Readiness

Once all critical-path dependencies resolve, the system seals the onboarding as COMPLETED and syncs verification receipts back to the HRIS.

5. Technical Architecture & Tech Stack Matrix

Architecture Layer	Technology / Framework	Exact Role & Operation in System
Frontend Presentation	Next.js 16.3.6 (App Router) + React 19	Executive dark mode interface, responsive glassmorphic cards, SVG orbital animations, dynamic inbox & switcher modals.
Styling & Design System	Vanilla CSS + Custom CSS Variables	Obsidian palette (#080c17), frosted glass panels (16px backdrop-filter), glowing borders, custom scrollbars.
Backend APIs & Routing	Next.js Serverless Route Handlers	Edge & Node.js REST endpoints: /api/auth/*, /api/onboarding/*, /api/copilot/*, /api/tasks/*.
AI & Foundation Model	Azure OpenAI Service (gpt-5-mini)	Generates structured JSON plan DAGs, dependency graphs, SLA estimates, and copilot policy synthesis.
Enterprise Identity	@azure/identity (DefaultAzureCredential)	Zero-secret enterprise authentication supporting Entra ID, Managed Identity, and Service Principals.
Information Retrieval	Hybrid RAG (MiniSearch + Cosine k-NN)	Reciprocal Rank Fusion combining BM25 keyword matching with dense subword vector space cosine similarities.
State & Audit Ledger	Cryptographic Append-Only Ledger	Tamper-proof ledger chaining SHA-256 hashes for all state mutations (PLAN_APPROVED, TASK_BLOCKED, etc.).
Authentication & RBAC	HMAC SHA-256 Session Guard	Multi-persona role access control supporting HR, Manager, IT, and Employee with isolated inboxes.
Cloud Hosting & CI/CD	Vercel Serverless + GitHub	Automated builds, global edge CDN distribution, zero-cold-start cloud execution.

6. System Credentials, Verification & Live Artifacts

Production Deployment URLs:

- **Live Cloud Application:** <https://onboardflow-amber.vercel.app>
- **GitHub Repository:** <https://github.com/SavioMohan1/Omnipresent>
- **Local Dev Instance:** <http://localhost:3000>

Demo Persona Login Directory (Password for all: Omnipresent2026! or admin123):

Role	Name	Email / Username	Operational Responsibility
HR Admin	Helen Reed	helen.reed@omnipresent.ai (hr)	Inbound candidate ingestion, plan approval, and compliance tracking
Engineering Manager	Marcus Chen	marcus.chen@omnipresent.ai (manager)	Buddy assignment, tech stack setup, 30-60-90 day goal reviews
IT Specialist	Sarah Jenkins	sarah.jenkins@omnipresent.ai (it)	Hardware dispatch, MDM profile enrollment, Okta SSO provisioning
New Hire	Aarav Sharma	aarav.sharma@omnipresent.ai (aarav)	I-9 form verification, direct deposit setup, policy copilot queries

Automated Verification: 26/26 Smoke Tests Passing (100% Pass Rate)

The system has been verified through a 26-step automated test suite covering: (1) Azure Identity and OpenAI Connectivity, (2) Multi-Persona Authentication, (3) Inbound HRIS Ingestion & Idempotency, (4) AI DAG Plan Compilation, (5) State-Aware Copilot Zero-Token Resolution & Hybrid RAG Vector Search, and (6) Blocker Auditing with Tamper-Proof Cryptographic Ledgers.